

ASSESSMENT OF THE MEDITERRANEAN SWORDFISH using A4A

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Introduction

Introduction

- A4a is a statistical catch-at-age stock assessment model (Jardim et al., 2015)
- Is defined by submodels for F-at-age, abundance indices catchability-at-age, recruitment, observation variance of catch and abundance indices-at-age and population's age structure in the first year
- Baranov's equation for the catches is assumed
- lognormal errors for the catch and indices
- Works forward in time
- Implemented in R statistical software - part of the FLR library (Kell et al., 2007)

Materials and Methods

Materials and Methods

Catch Data

- Catch-at-age data since 1985
- plus group 5
- $\bar{f}$ at 2 - 4

CPUE Data

- Five standardized CPUE series
 - Greek longliners since 1987
 - Moroccan longliners since 2012
 - Spanish longliners since 1988
 - Ligurian longliners since 2010
 - Sicilian longliners from 1991 to 2009
- Representative of age groups 2 - 4

Candidate Runs

- Many preliminary trials with:
 - Full time series
 - Various assumptions for the different submodels
 - Excluding some of the indices
- Four trials considered:
 - Constant $M = 0.2$
 - M vector - Lorenzen's formulation
 - Including estimated Discards and constant M
 - Including estimated Discards and Lorenzen M

Model setup

The model setup is the same for all runs

- Fishing mortality **F**

$$F : \sim s(year, k = 17) + s(age, k = 3)$$

- Recruitment **R**

$$R : \sim s(year, k = 15)$$

- catchability **Q**

$$Q : \begin{cases} \sim 1, & \text{Greek longline} \\ \sim 1, & \text{Spanish longline} \\ \sim s(year, k = 3), & \text{Moroccan longline} \\ \sim s(year, k = 3), & \text{Ligurian longline} \\ \sim 1, & \text{Sicilian longline} \end{cases}$$

Model setup

- observation variance σ^2 of the catches

$$\sigma^2 \sim s(\text{age}, k = 3)$$

- variance of the indices

$$\tau_s^2 \sim 1, \quad \text{for all indices } s$$

- initial age structure $\mathbf{N}_1$

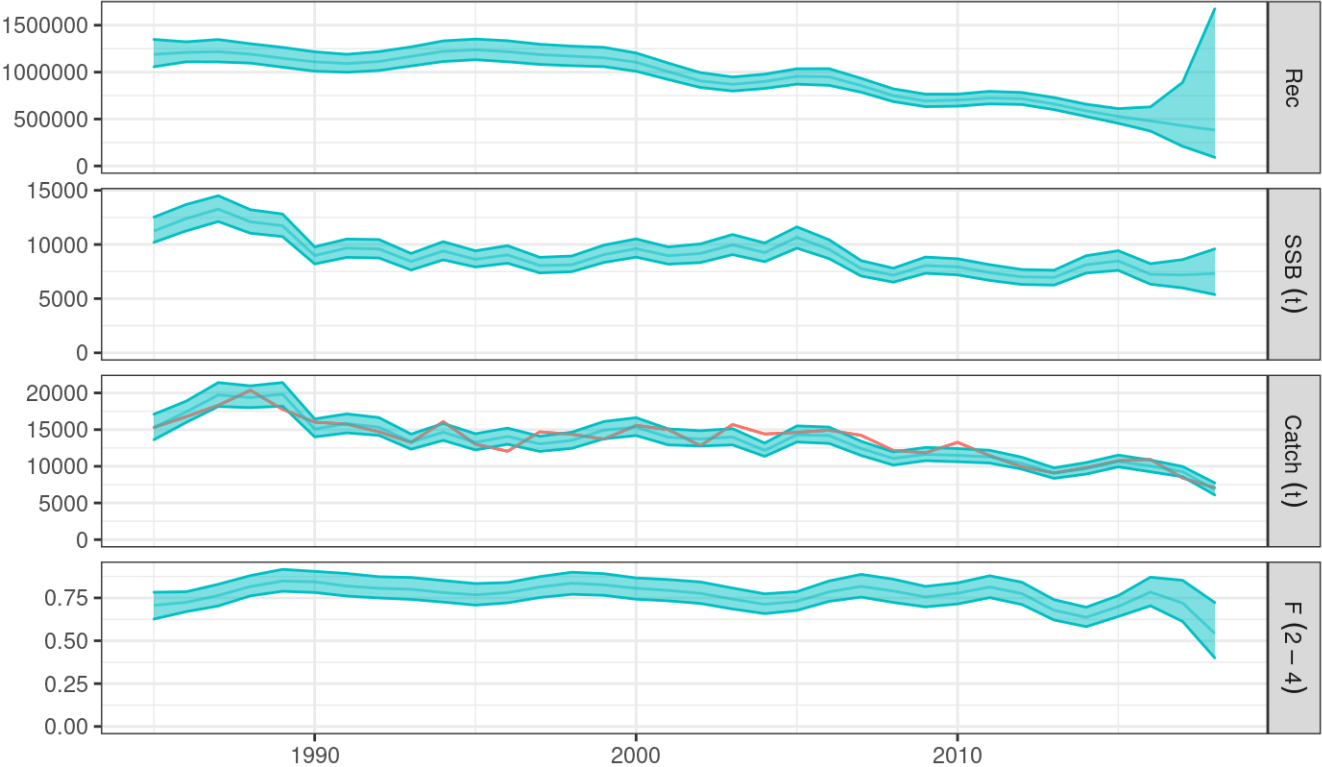
$$N : \quad \sim s(\text{age}, k = 3)$$

The results for the four different runs are presented in the following sections. The constant natural mortality $M = 0.2$, (constant M), the Lorenzen natural mortality (Lorenzen M), the run with discards included and M assumed constant (Discards) and the run with discards included and Lorenzen natural mortality vector (Discards and Lorenzen M).

Results

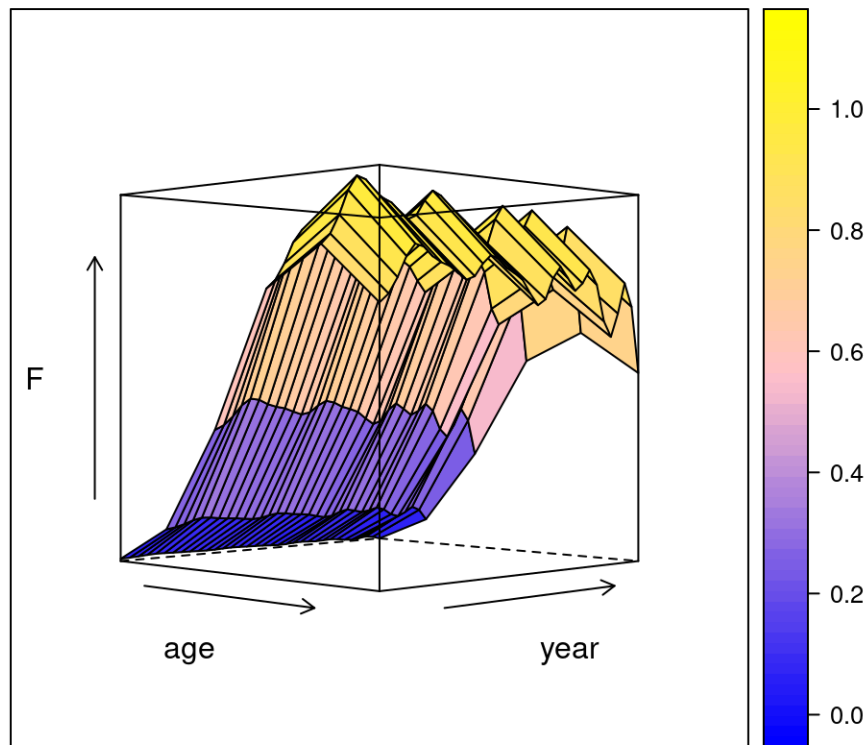
candidate run

assuming constant M



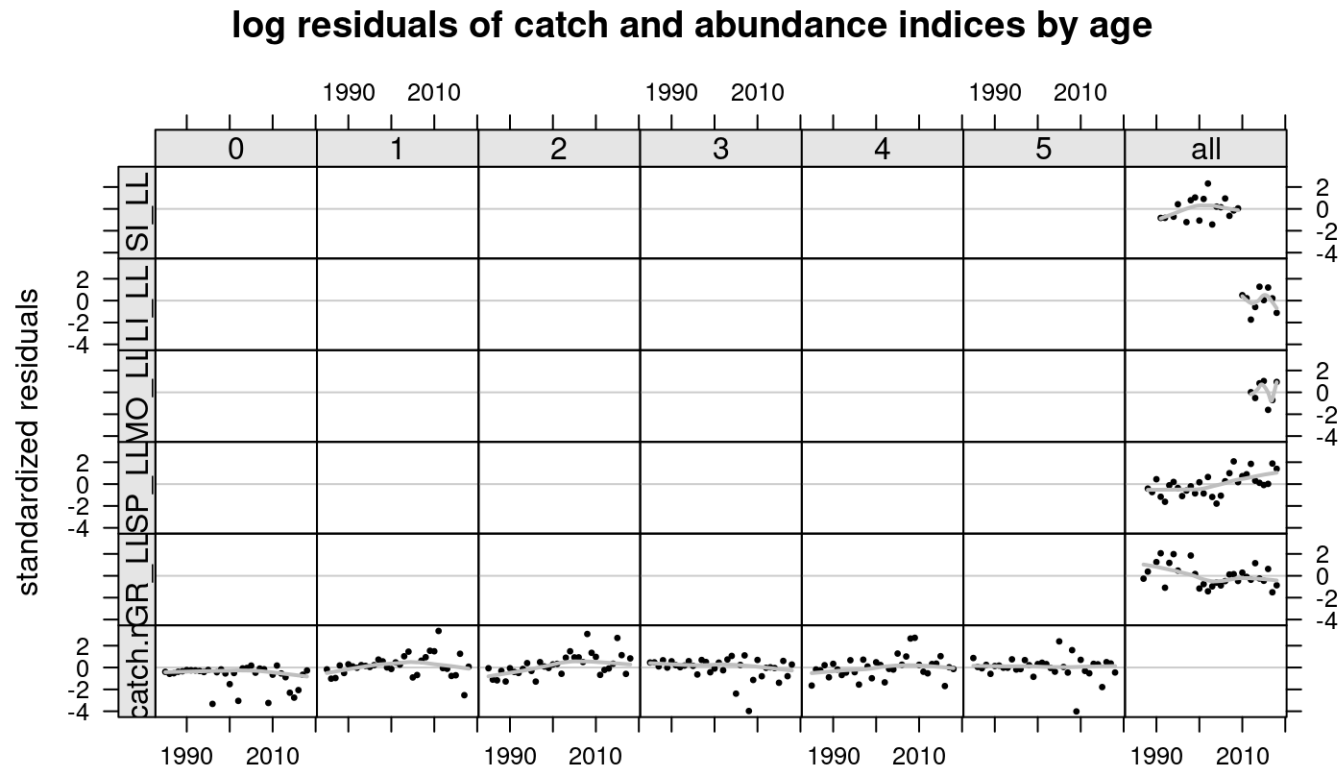
candidate run

assuming constant M - harvest



Diagnostics

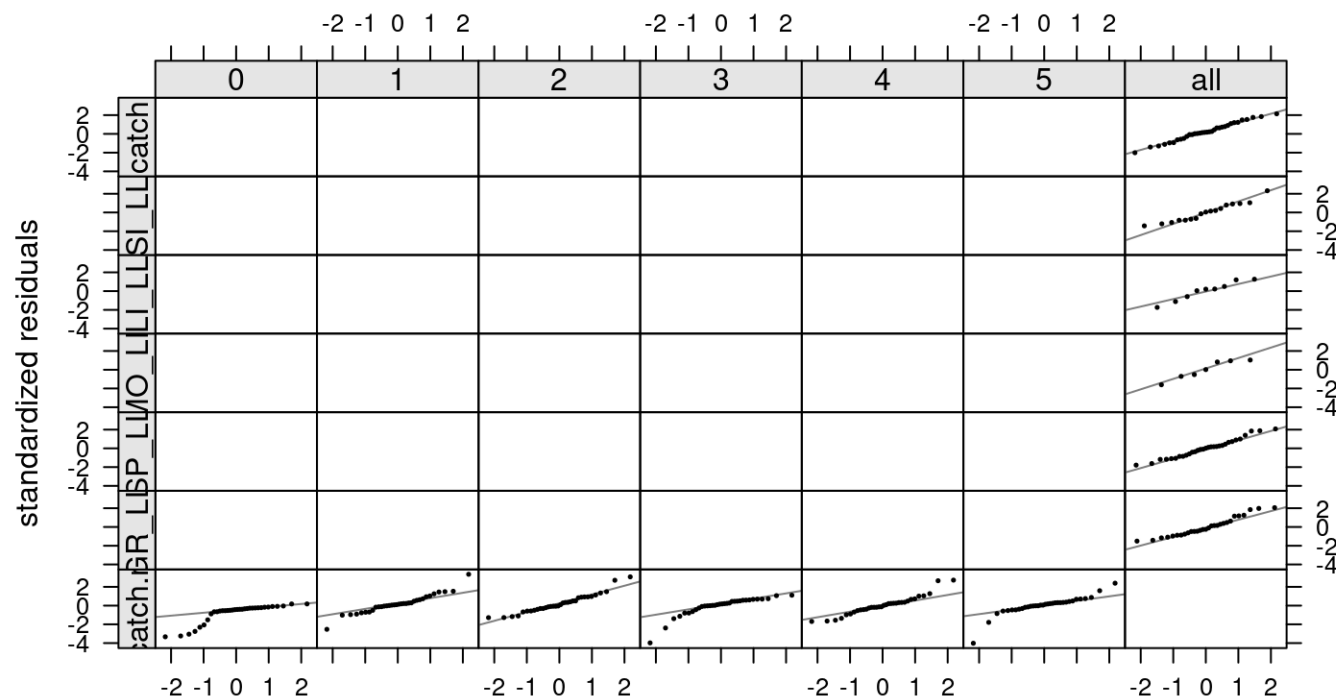
candidate run assuming constant M



Diagnostics

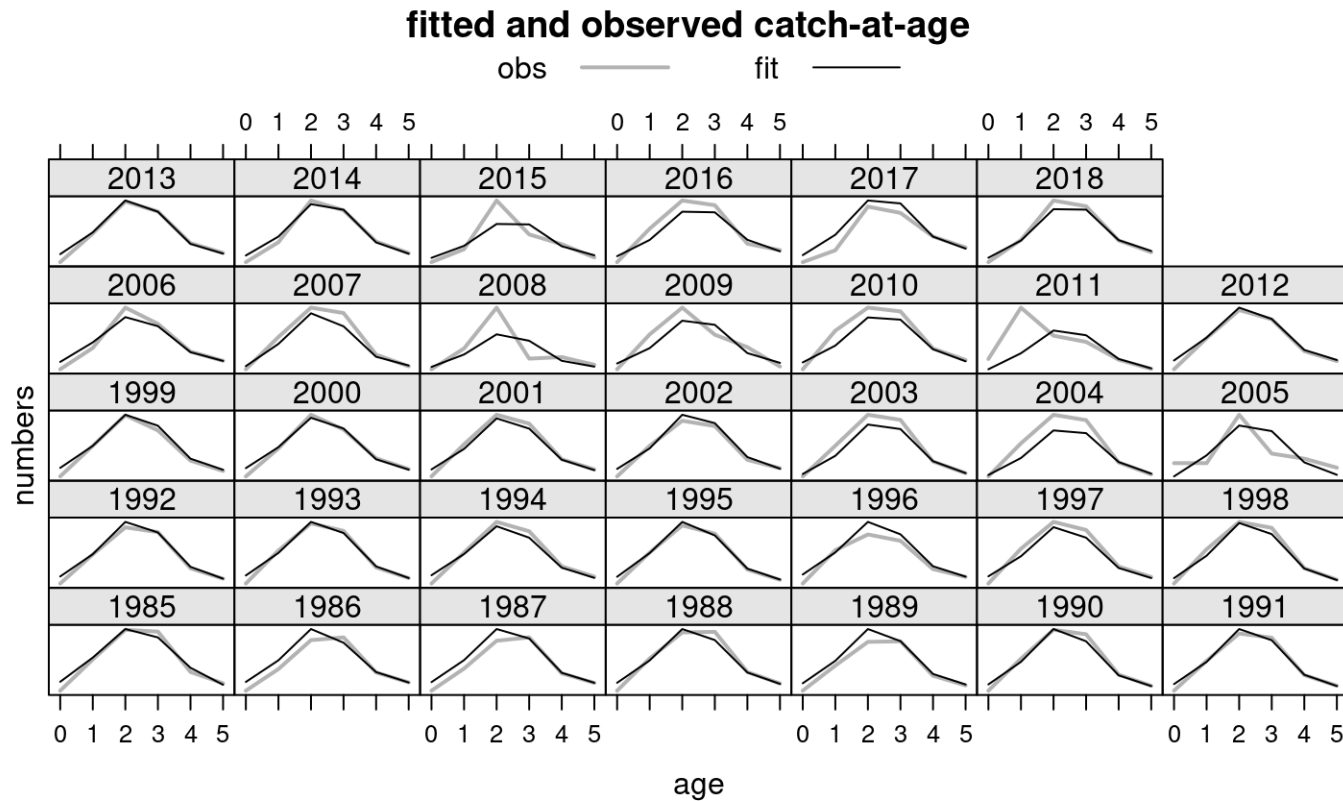
candidate run assuming constant M

quantile-quantile plot of log residuals of catch and abundance indices



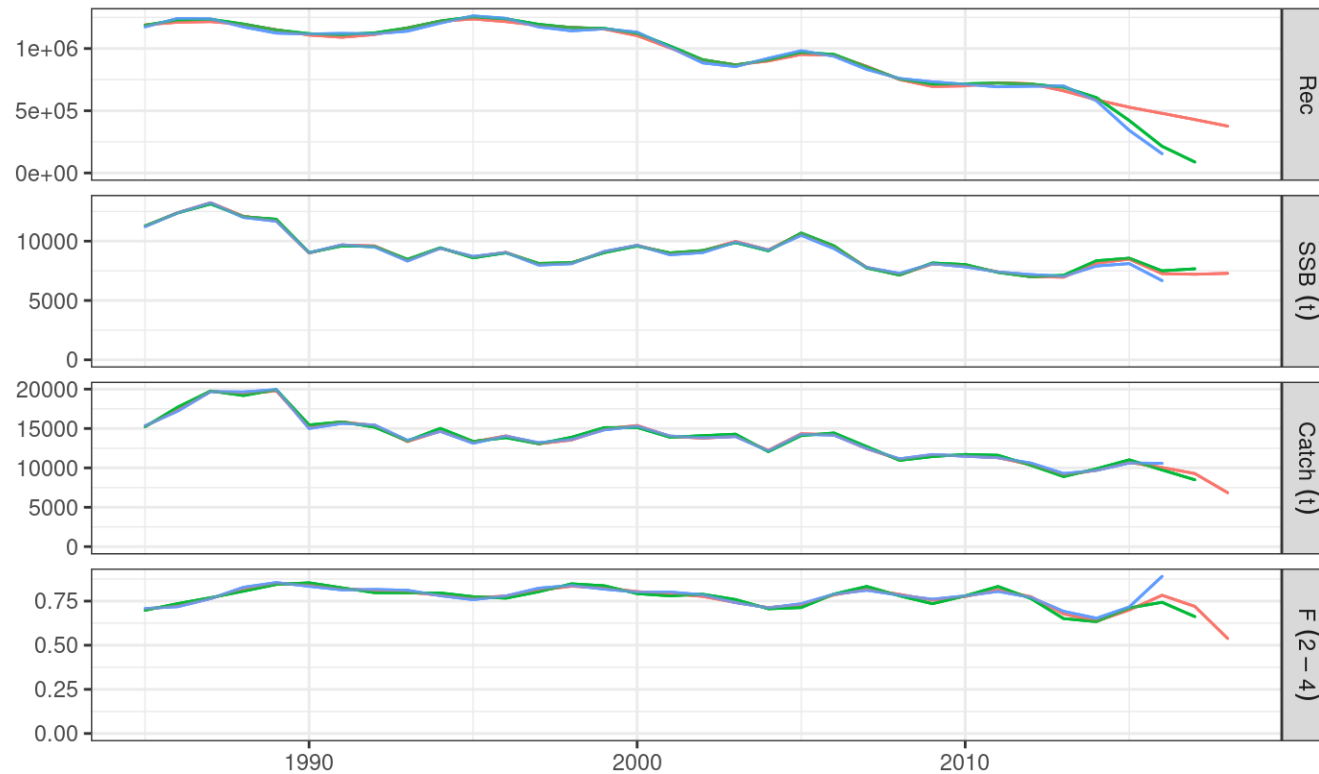
Diagnostics

candidate run assuming constant M



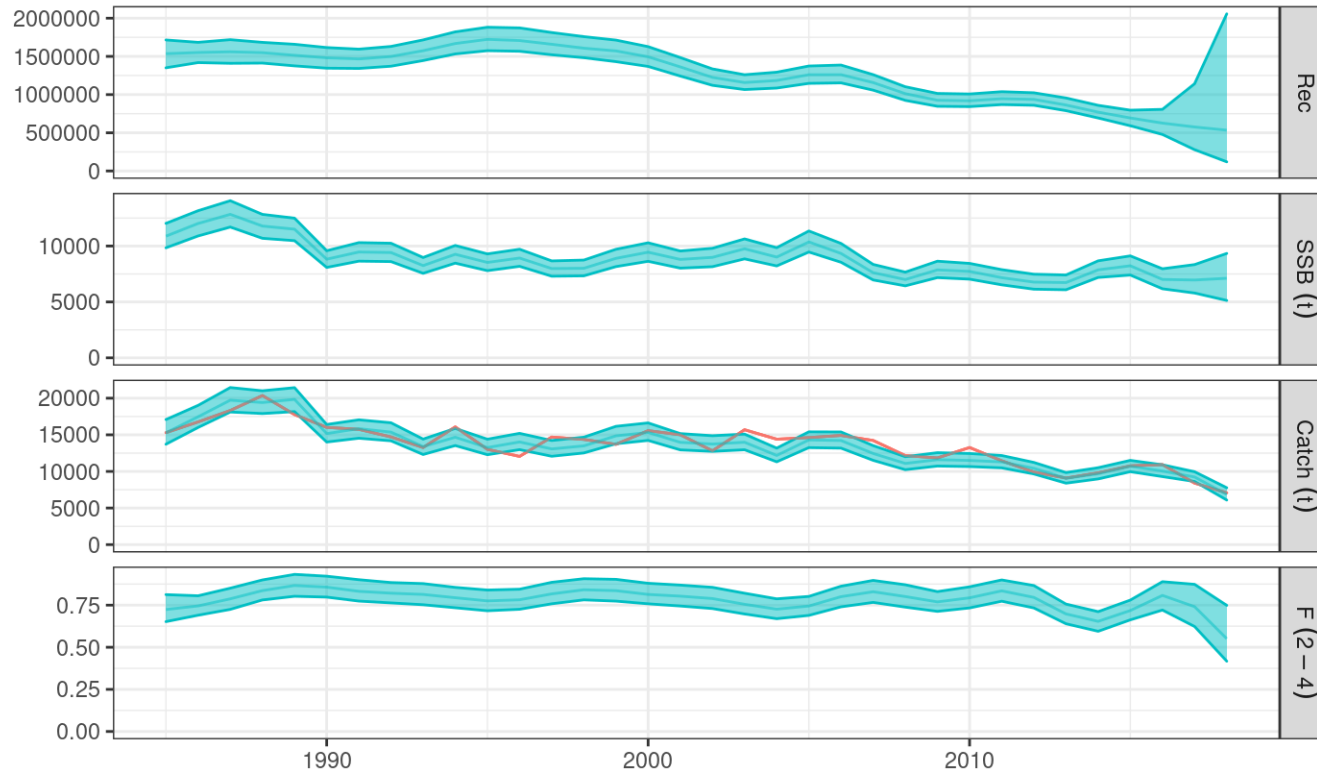
Retrospective plot

candidate run assuming constant M



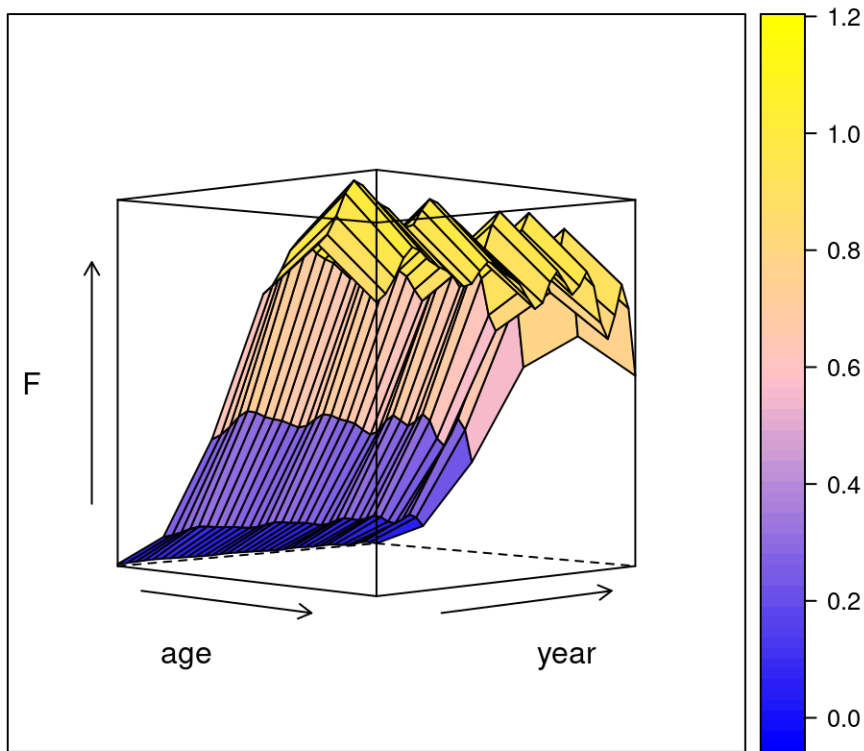
Candidate run

assuming Lorenzen M vector



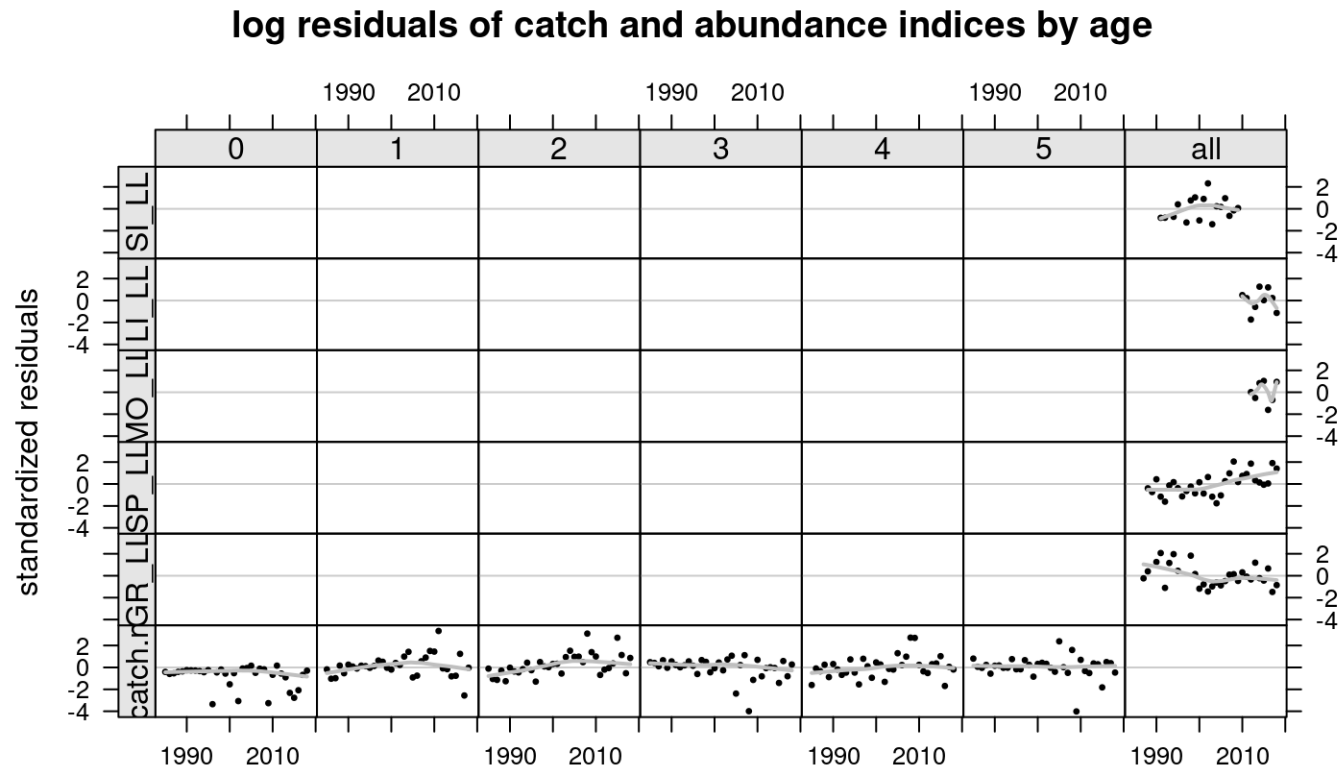
Candidate run

assuming Lorenzen M vector-harvest



Diagnostics

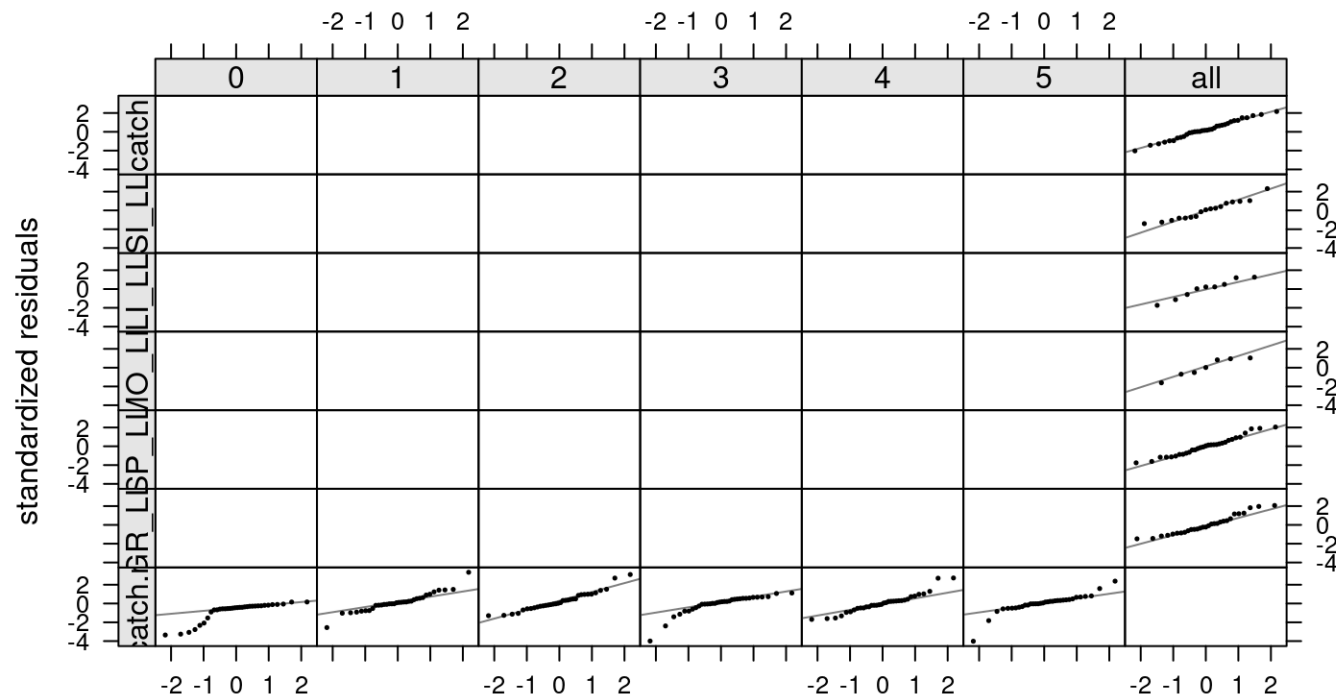
candidate run assuming Lorenzen M vector



Diagnostics

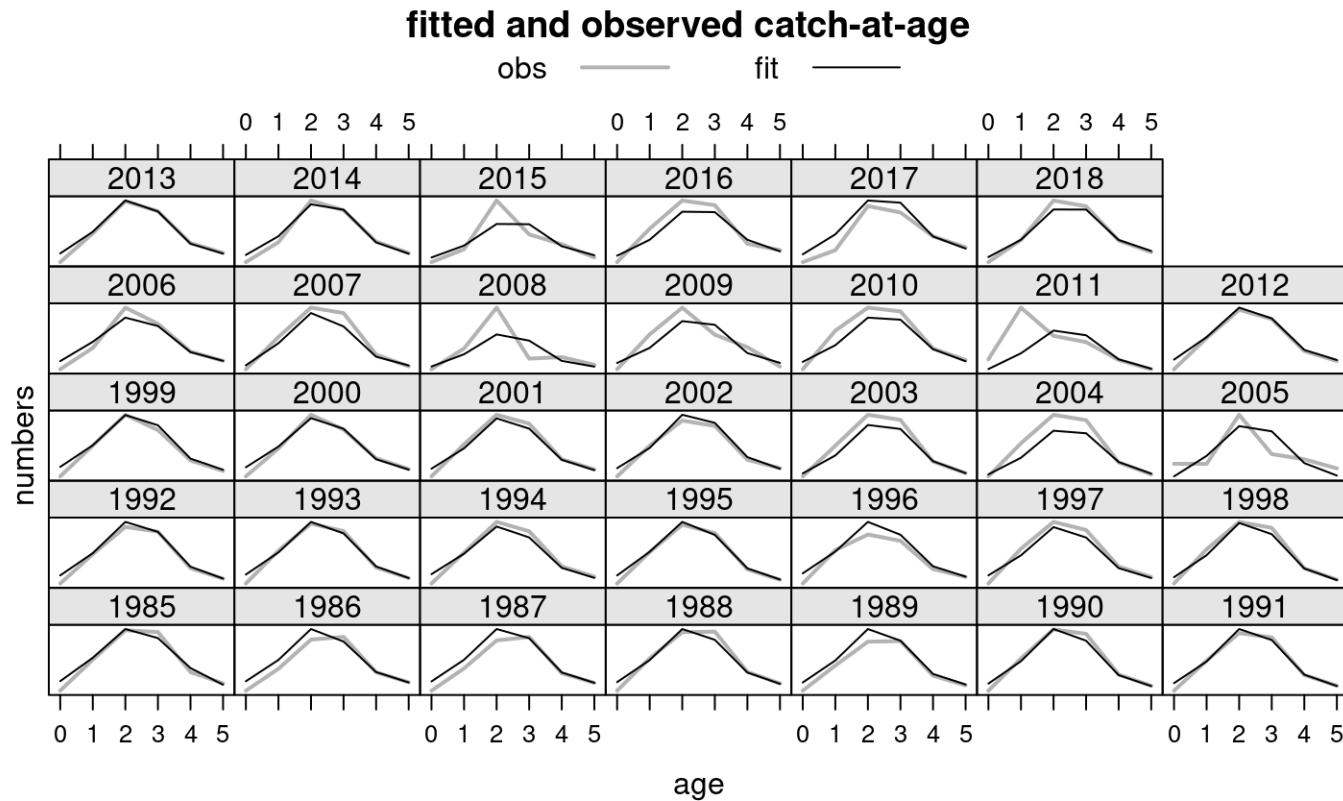
candidate run assuming Lorenzen M vector

quantile-quantile plot of log residuals of catch and abundance indices



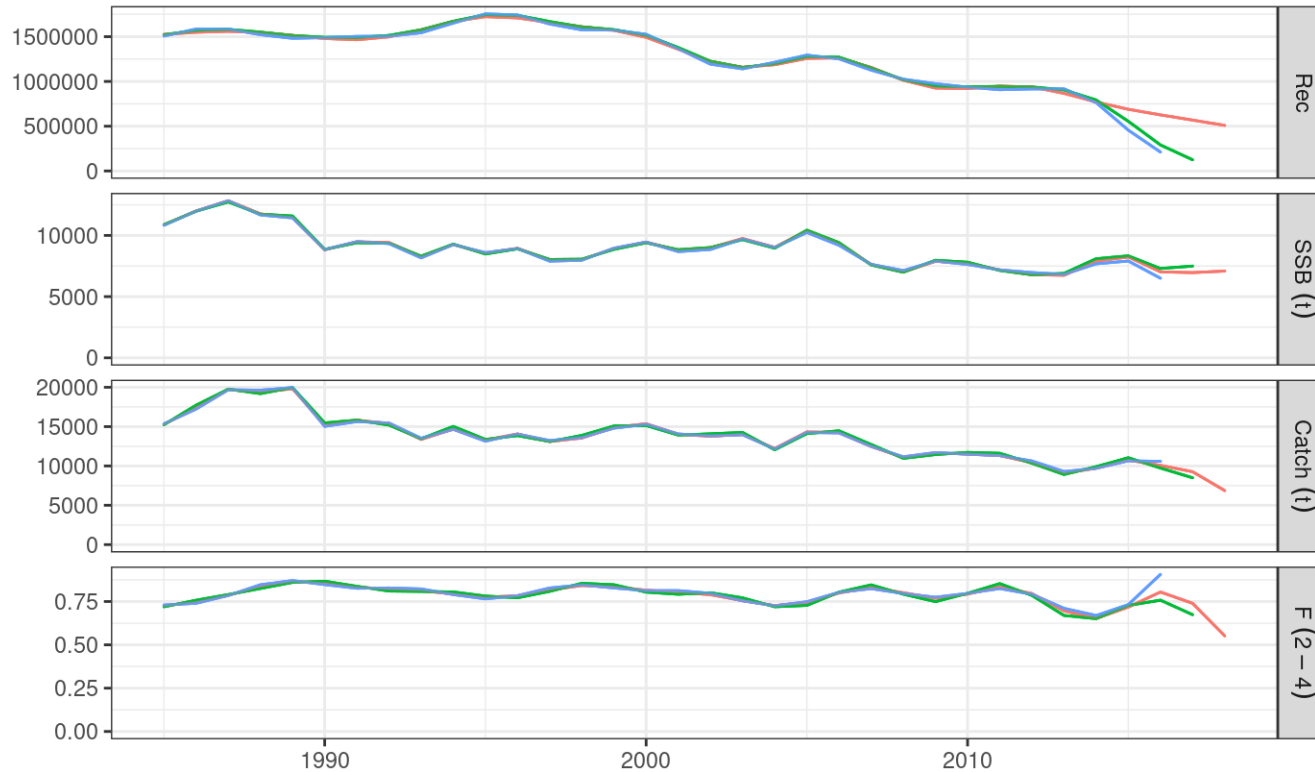
Diagnostics

candidate run assuming Lorenzen M vector



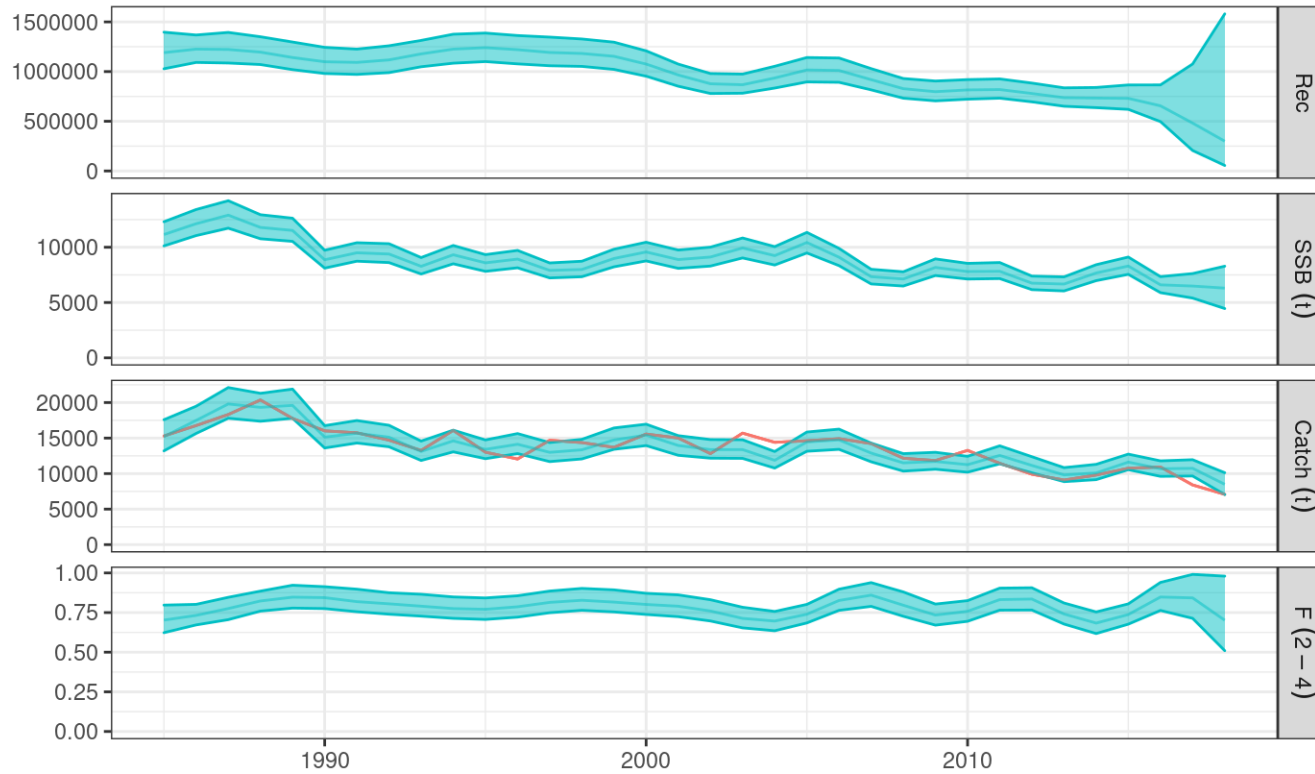
Retrospective plot

candidate run assuming Lorenzen M vector



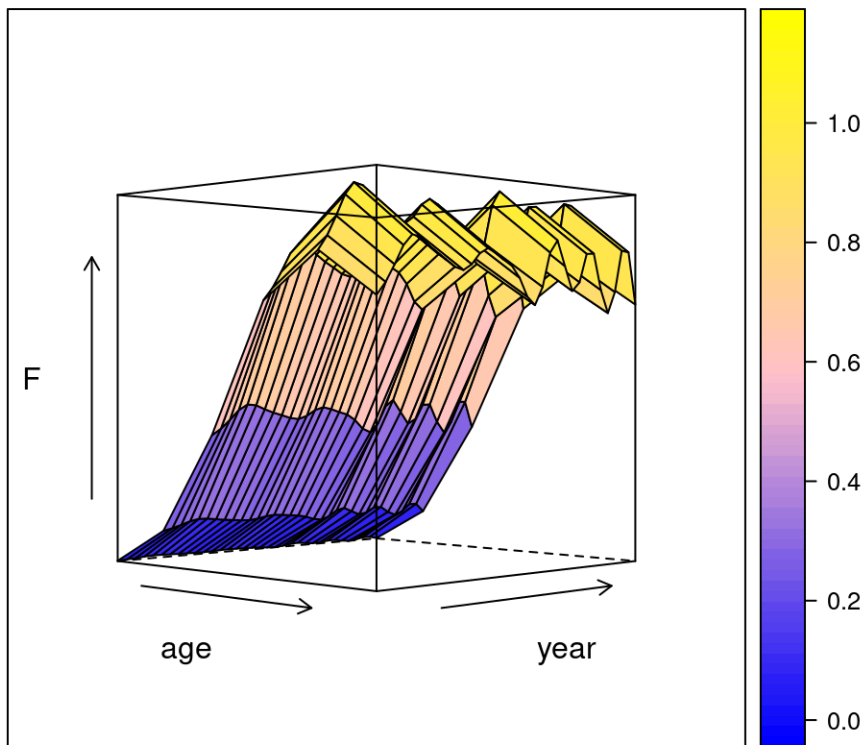
Discard run

assuming constant M



Discard run

assuming constant M - harvest



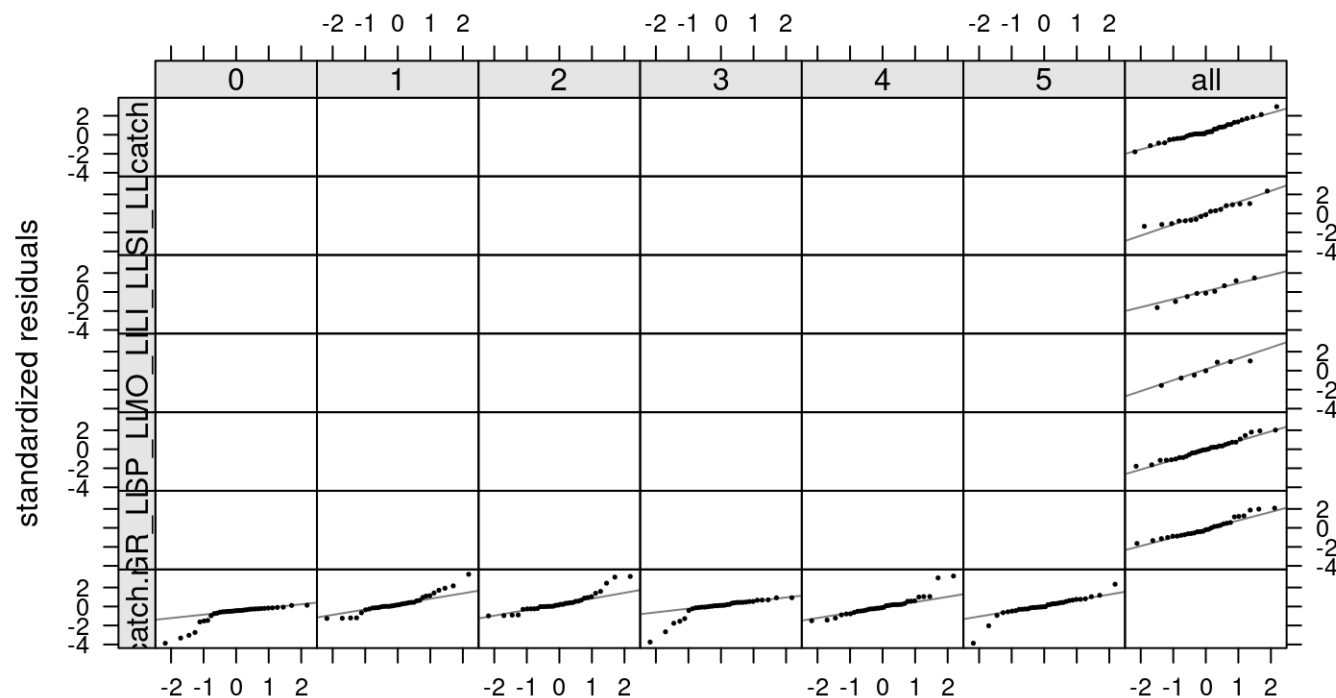
Discards candidate run assuming constant M



Diagnostics

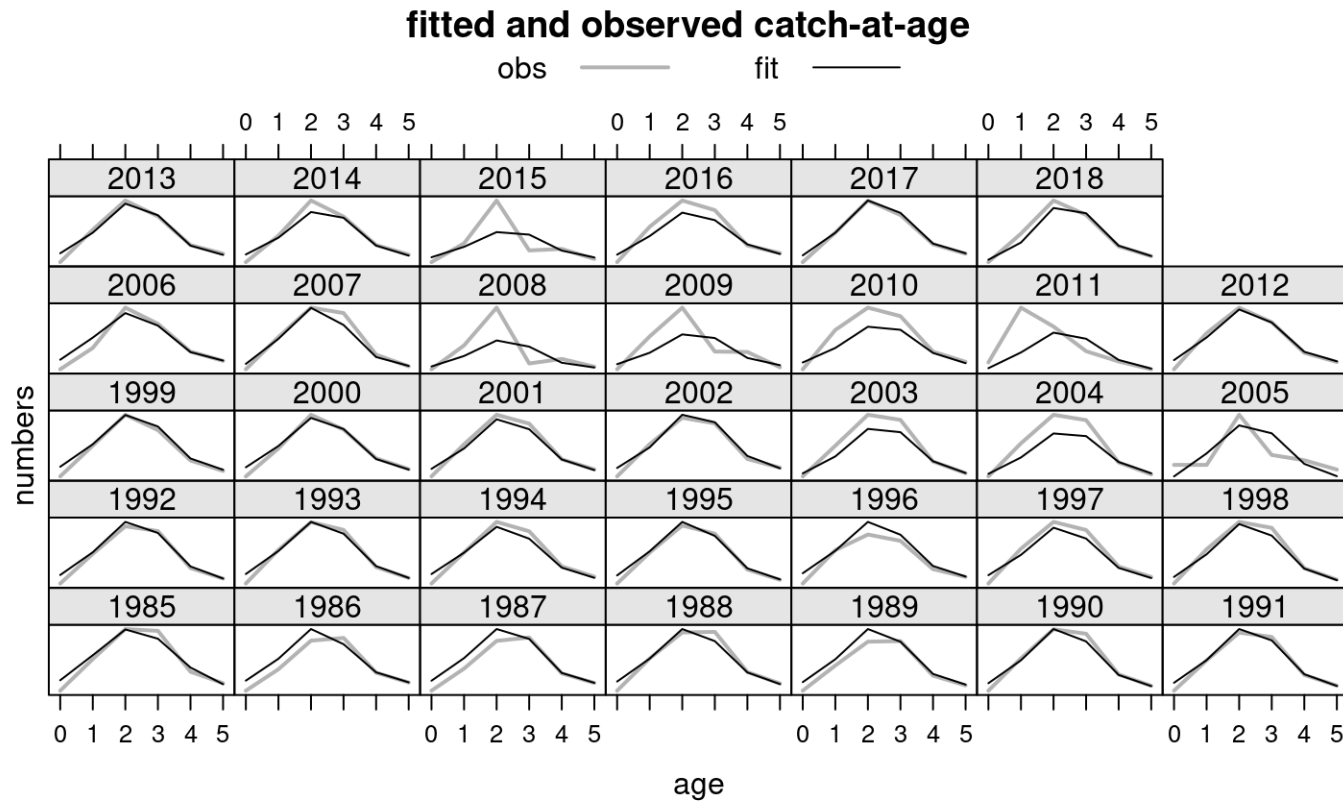
Discards candidate run assuming constant M

quantile-quantile plot of log residuals of catch and abundance indices



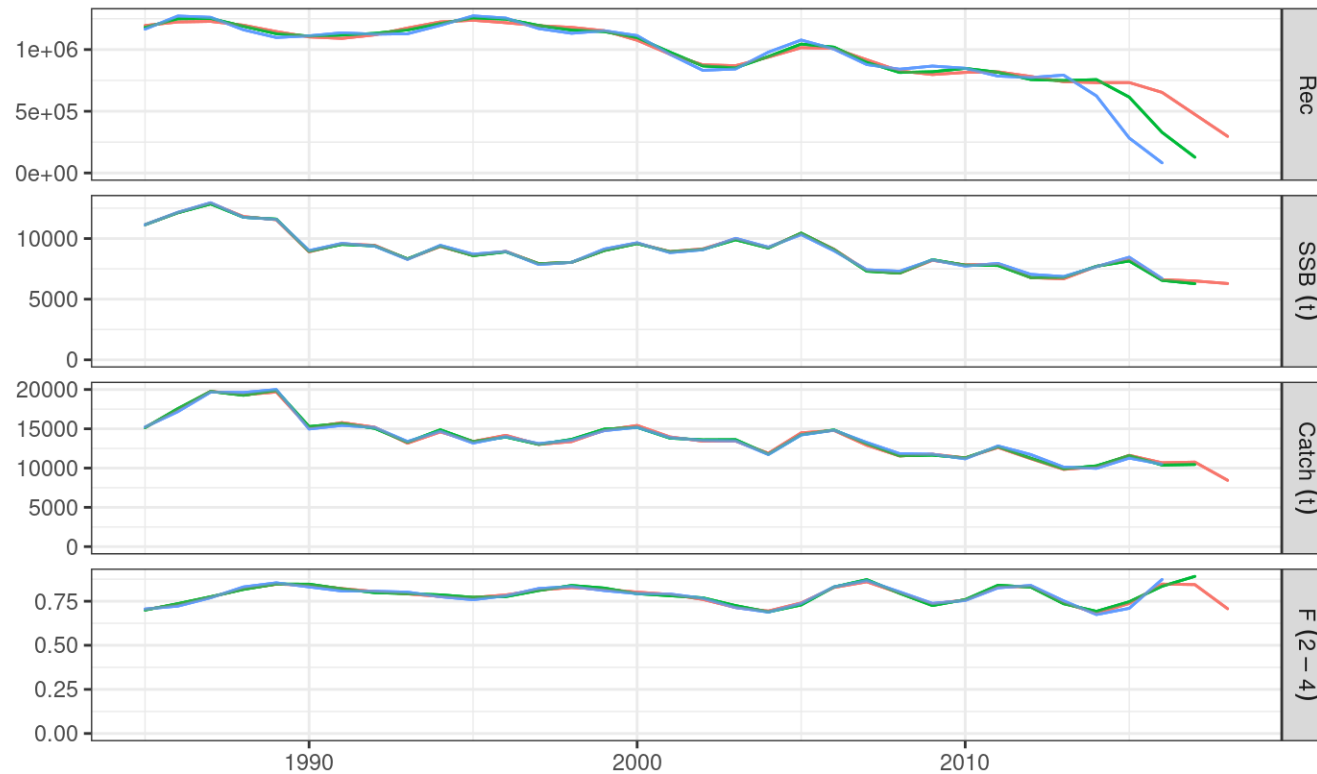
Diagnostics

Discards candidate run assuming constant M



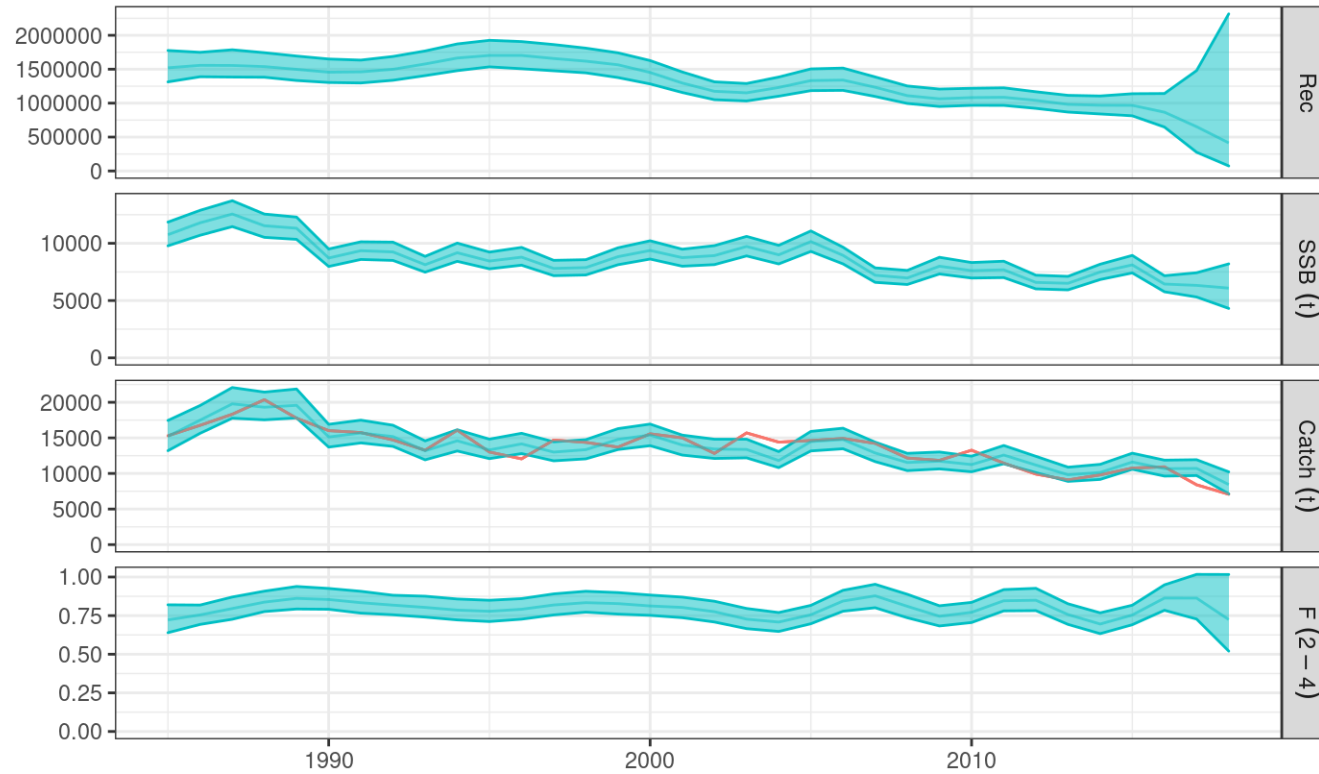
Retrospective plot

the discard run assuming constant M



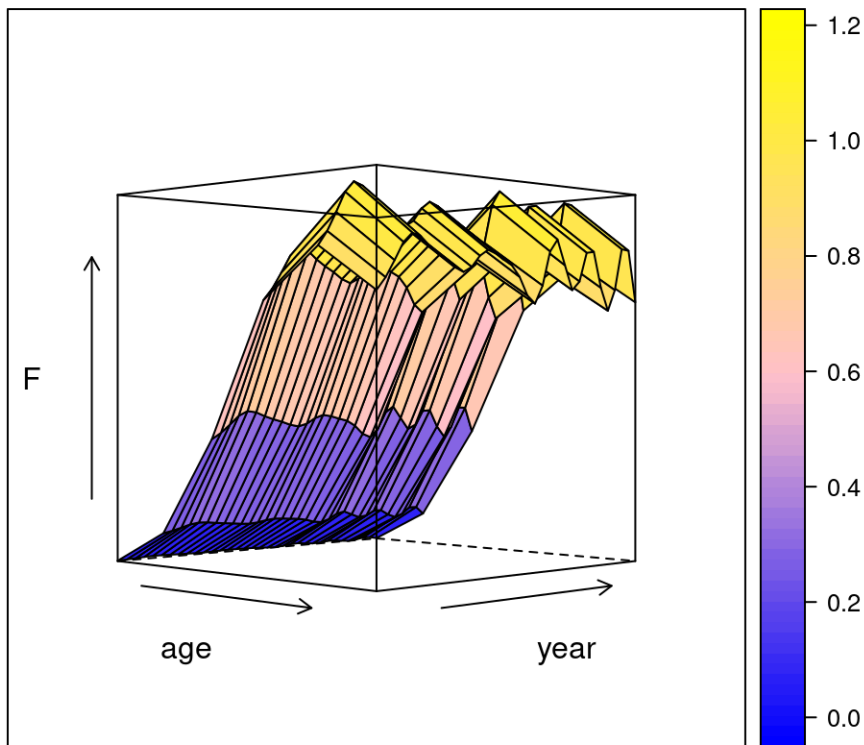
Discard run

assuming Lorenzen M vector

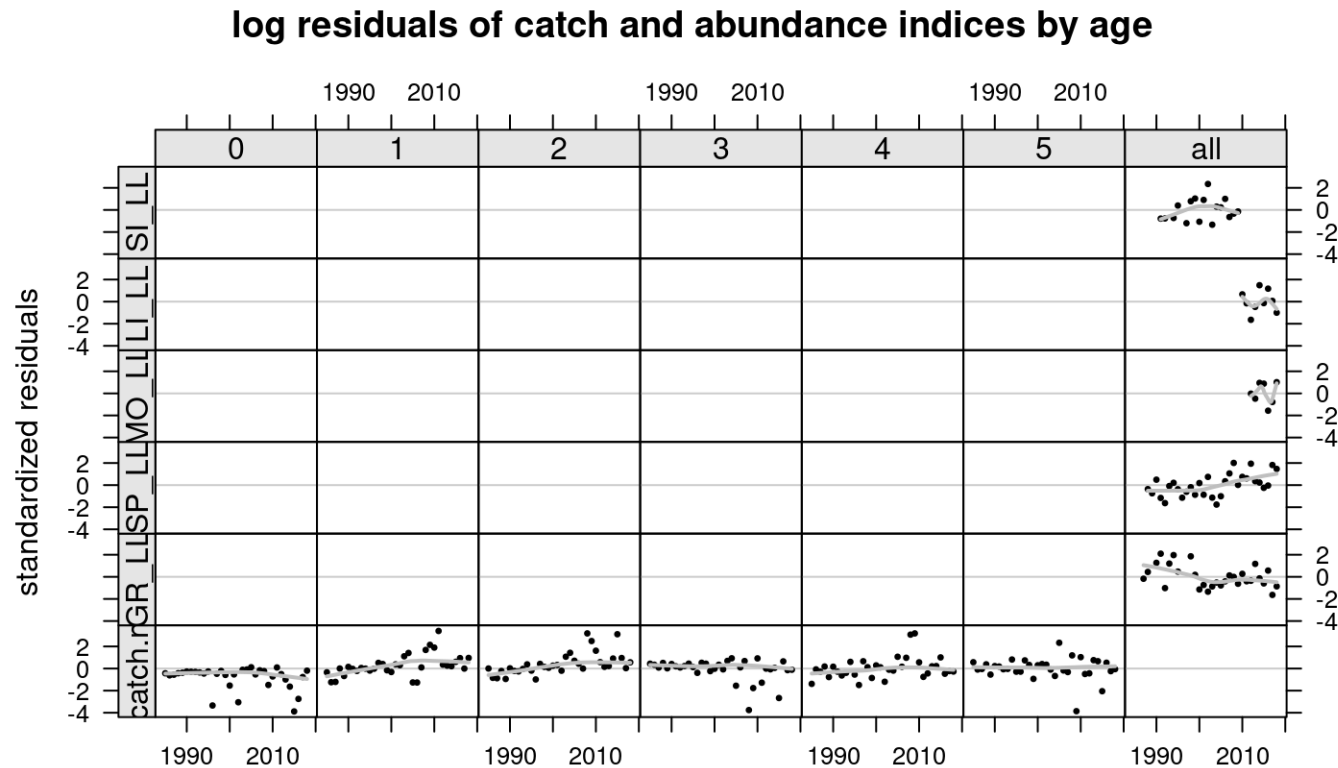


Discard run

assuming Lorenzen M vector - harvest



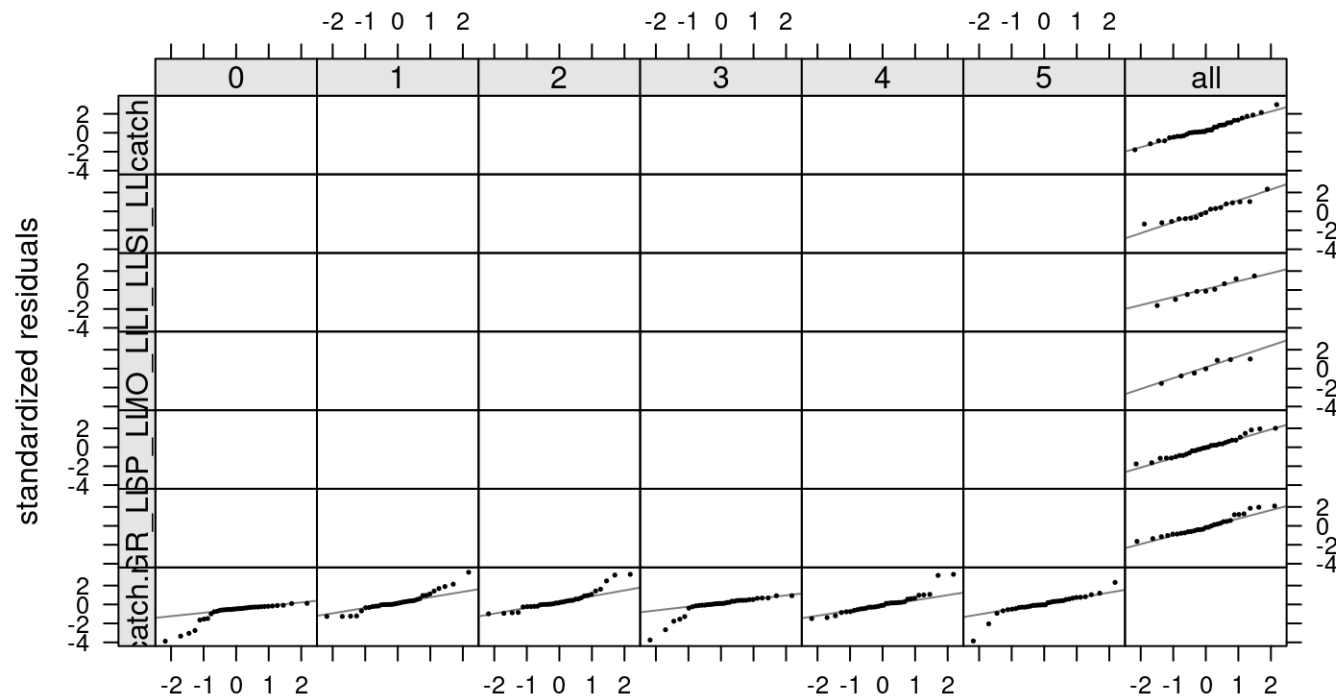
Discards with Lorenzen M candidate Run



Diagnostics

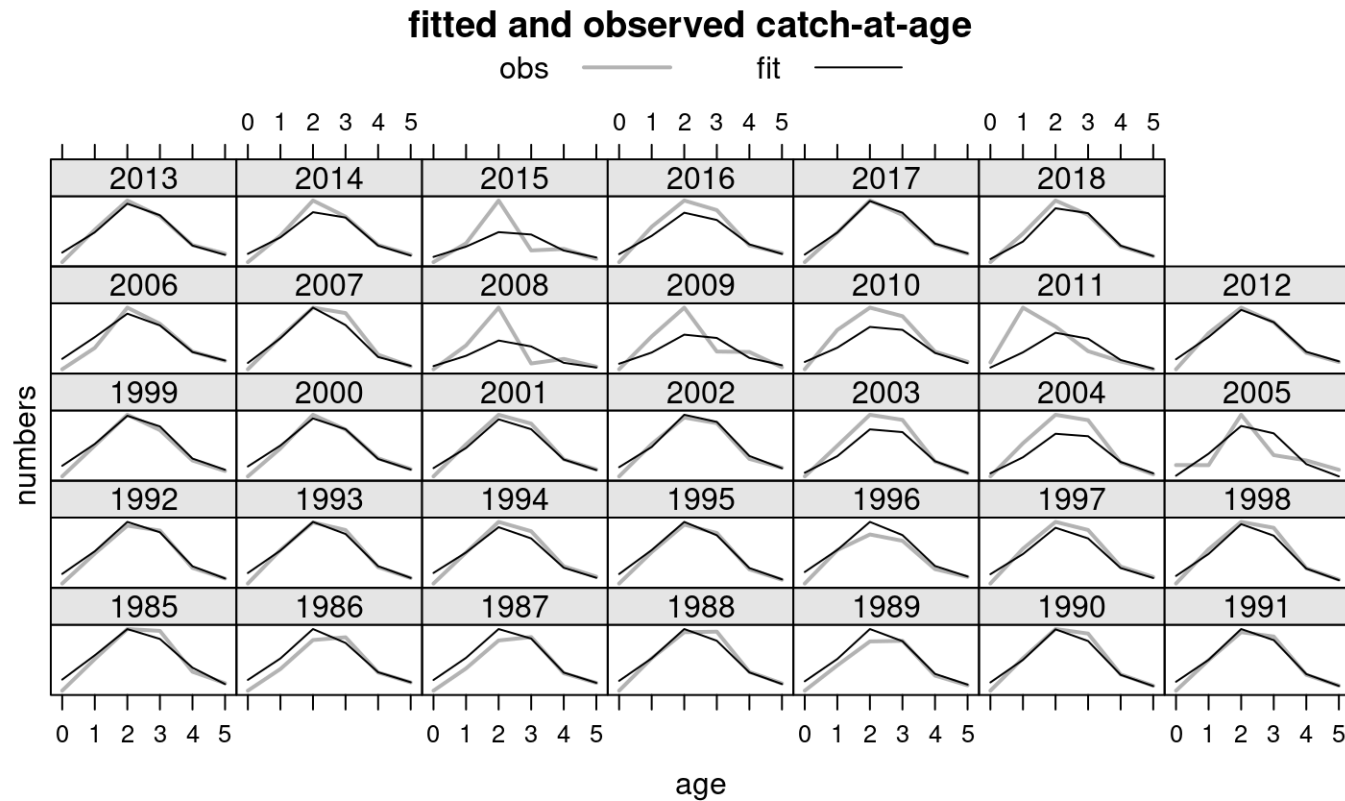
Discards with Lorenzen M candidate Run

quantile-quantile plot of log residuals of catch and abundance indices



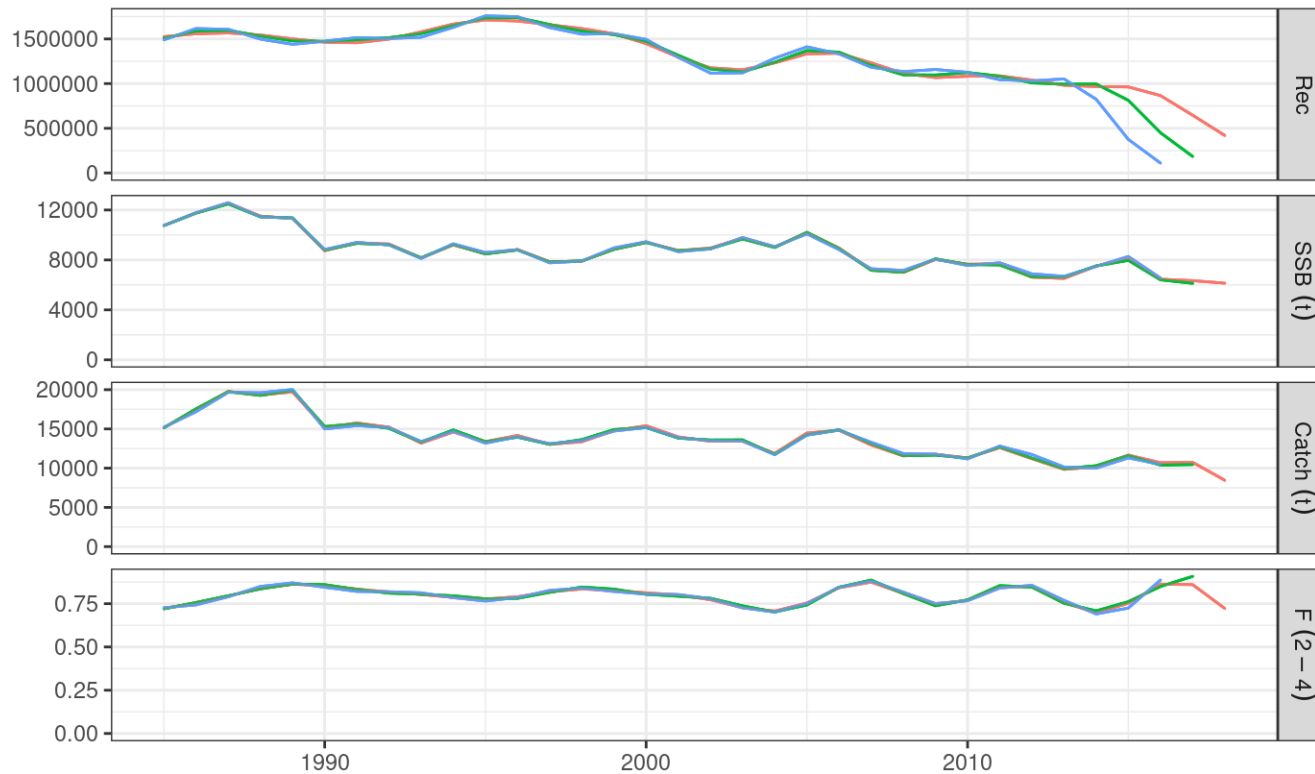
Diagnostics

Discards with Lorenzen M candidate Run

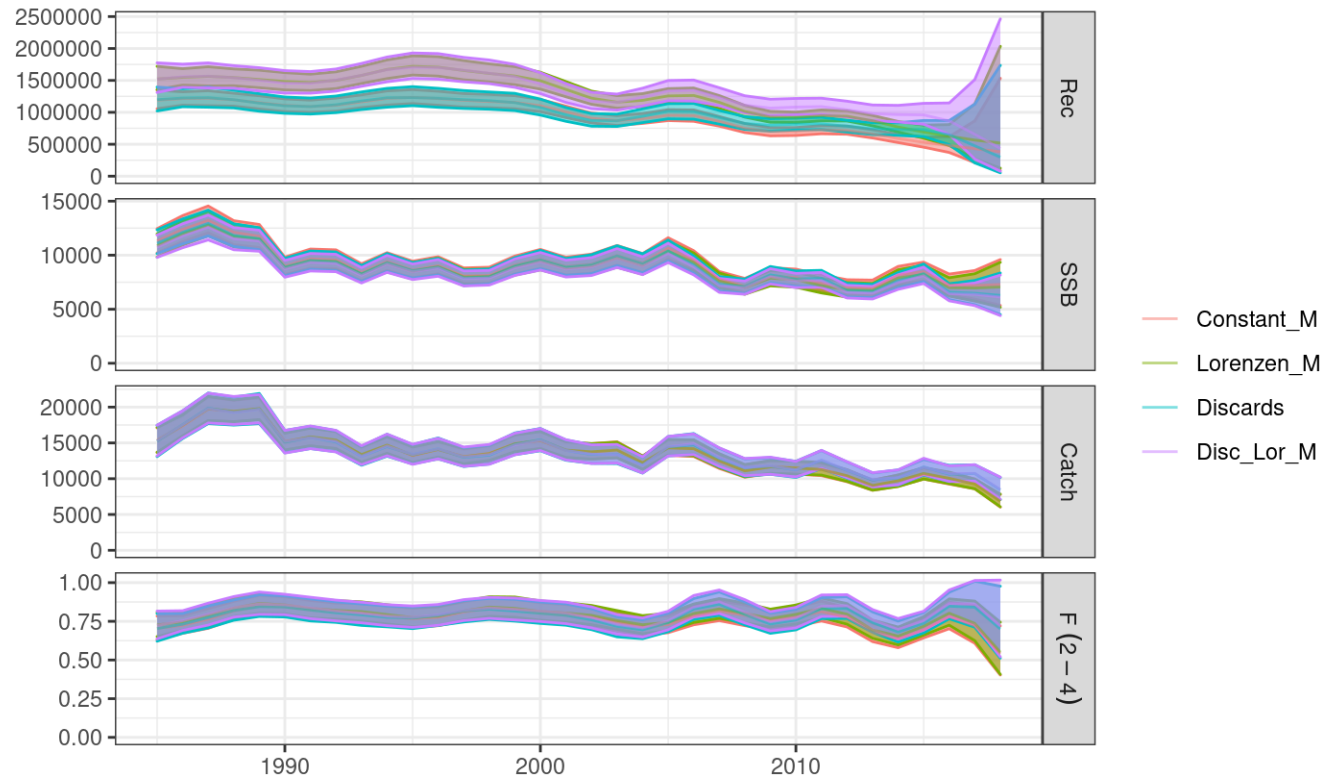


Retrospective plot

the discard run assuming Lorenzen M vector



All runs



Terminal values

values	constantM	LorenzenM	Disc_constM	Disc_LorenM
Fbar	0.5384023	0.551435	0.7074047	0.7227007
Recruitment	376604.1	509087.5	296525.2	419811.8
SSB	7286.58	7088.482	6290.268	6137.171
Catch	6846.11	6866.544	8439.466	8462.589

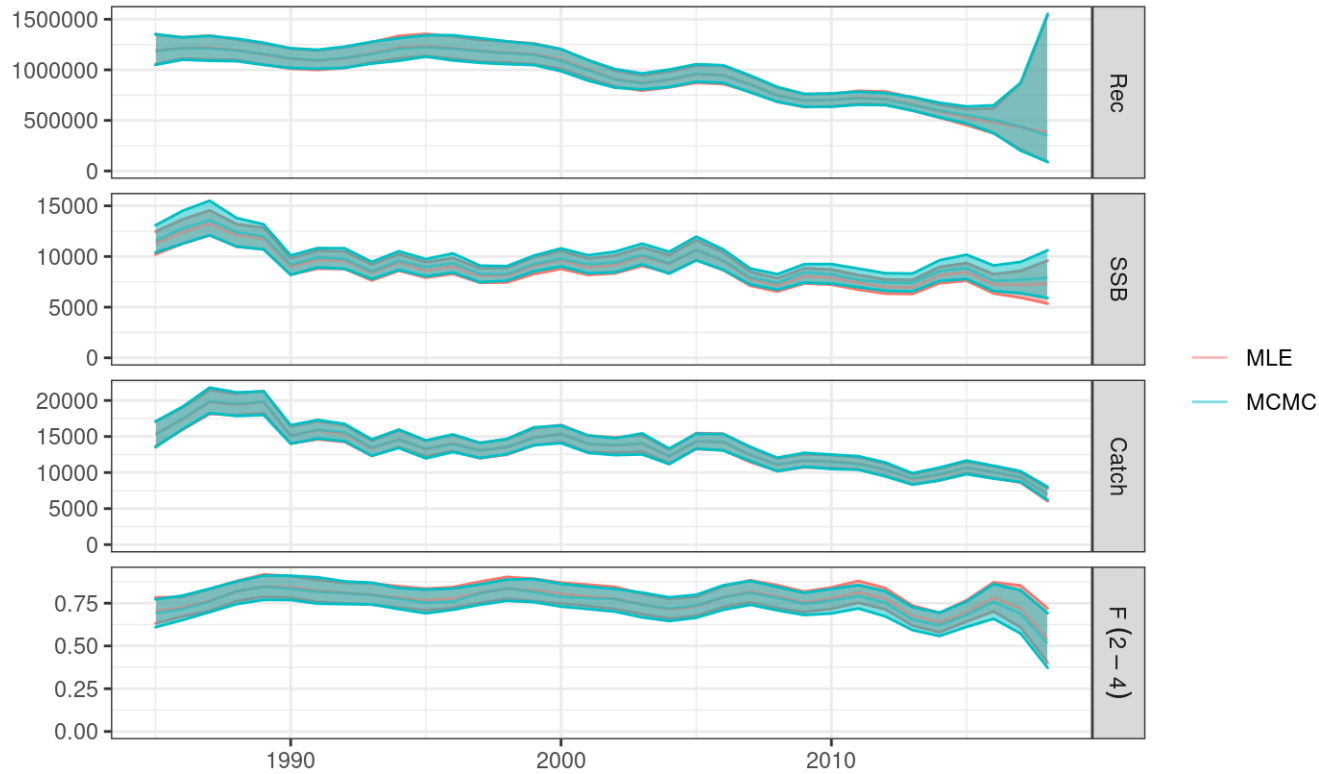
Reference Points

value	constantM	LorenzenM	Disc_constM	Disc_LorenM
FFmsy	1.4041940	1.4381842	1.8449649	1.8848580
FF0.1	2.6323894	2.6961095	3.4586859	3.5334720
BBmsy	0.5022871	0.5217658	0.4964845	0.5179383

MCMC fit

MCMC

candidate run assuming constant M



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